RAJALAKSHMI ENGINEERING COLLEGE
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(Ministry of HRD Initiative)

Centre For Design Thinking

PROUDLY PRESENTS

4th DESIGN THINKING PROJECT CONTEST

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The Stealth-Fold: Space-Optimized Rear Passenger Utility

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ABSTRACT



2. The Goal: Engineering "Invisible" Utility

- **Objective:** To engineer a "Stealth-Fold" cup holder that integrates directly into the car's door trim, providing high-load support without sacrificing cabin space.
- **Design Requirements:** The solution must be rugged enough to withstand the mechanical shock of door slams, compact enough to maintain a flush profile, and user-friendly for all age groups.
- **Philosophy:** Applying Design Thinking principles to transform a secondary storage area into a primary ergonomic feature.

1. Problem Statement: The C ience Gap

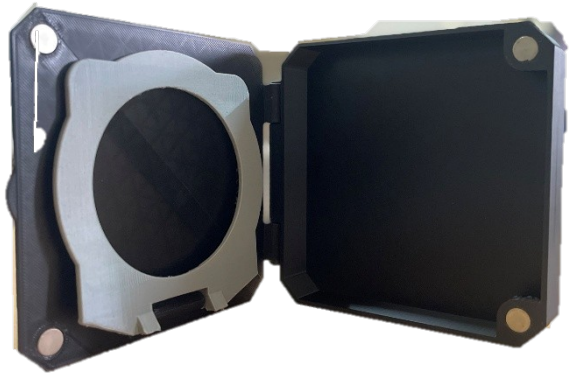
- **The Issue:** Modern comp vehicles prioritize seating capacit nger ergonomics, often negle age storage.
- **User Pain Points:** Rear p frequently forced to hold d nands or laps, leading to increased risk of spills, driver distraction, and damage to the vehicle's interior upholstery.

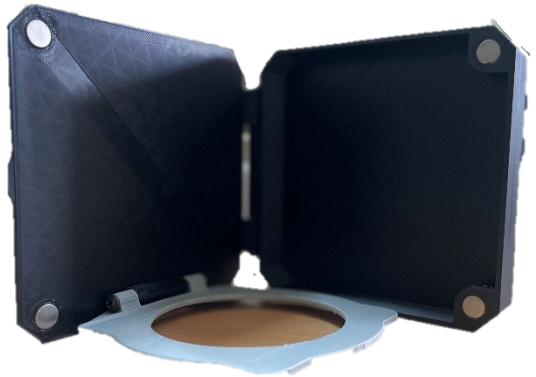
PROPOSED SOLUTION

An **integrated, "Stealth-Fold" cup holder** featuring a **magnetic-latch** for a flush, rattle-free, and rugged storag solution.

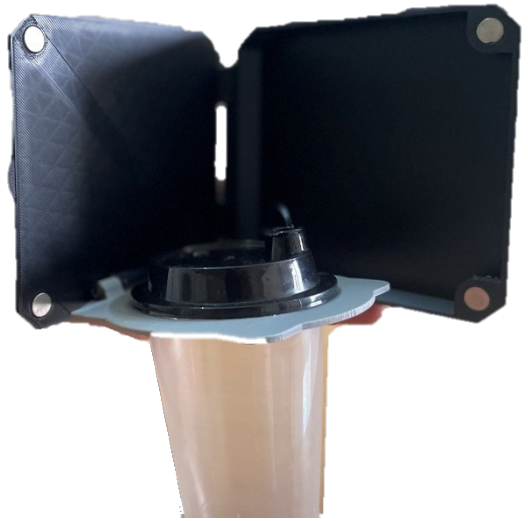


Stowed Position





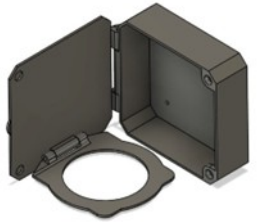
Deployment



Active Use

TECHNOLOGY USED

- **Design & Modeling Software (SolidWorks):** Used to develop the 3D parametric model of the cup holder assembly. This allowed for precise tolerance testing of the hinge and magnetic latch to ensure a seamless flush-fit when retracted.
- **Slicing Engine (UltiMaker Cura):** Used to translate the CAD model into G-code for manufacturing. Settings were optimized for **high infill density** and **wall thickness** to ensure the prototype could withstand the weight of a full beverage.
- **Manufacturing Hardware (FDM 3D Printer):** Utilizing **Fused Deposition Modeling** enabled rapid prototyping, allowing the team to test the physical ergonomics and magne[illegible]th in real-time.



Material Strategy (Current & Future):

- **Prototyping Phase (PLA):** Chosen for its dimensional accuracy and ease of printing during the design iteration phase.
- **Production Goal (ASA/ABS):** The final product is intended to be manufactured using **ASA (Acrylonitrile Styrene Acrylate)** or **ABS**. These materials are automotive-grade standards because they offer high UV resistance and thermal stability, preventing warping in car interiors that reach high temperatures.

NOVELTY

1. Structural Superiority: In-built vs. Add-on

- **The Flaw of Add-ons:** Traditional aftermarket solutions often rely on fragile clips that attach to AC vents or window seals, which are not designed to bear weight and frequently snap under the load of a full beverage.
- **Integrated Design:** Our solution is engineered to be an **integral component of the door trim architecture**. By being "in-built," it leverages the structural integrity of the door panel itself, providing a much higher load-bearing capacity and a seamless, factor[illegible]s



2. Advanced Ergonomics: Passenger-Centric Engineering Strategic Positioning:

Unlike centre console holders that require passengers to lean forward or floor-mounted versions that are hard to reach, this design is placed within the "natural reach zone" of a seated rear passenger.

Zero-Interference Layout: The compact, "Stealth-Fold" footprint is specifically mapped to avoid obstructing critical door functions like window switches or door release handles. This ensures that utility does not come at the cost of safety or existing

SOCIAL RELEVANCE

- **Enhancing Road Safety & Minimizing Distraction**
- **Promoting Hygiene & Interior Longevity**
- **Inclusive Comfort & Universal Design:**



REFERENCES

Mosaic Manufacturing. (2025). *ABS vs ASA - Mosaic Manufacturing*. https://mosaicmfg.com/abs-vs-asa/

- *Focus:* Detailed comparison of 3D printing materials, establishing ASA as the superior choice for automotive interiors due to its UV resistance and thermal stability.

Kashyap, S., & Ruhela, S. (2024). *Design thinking in automotive design*. Scribd. https://www.scribd.com/presentation/561715551/sics

- *Focus:* Case studies applying Design Thinking (Empathize, Define, Ideate, Prototype, Test) specifically to automotive interior innovations.

THANK YOU